

Requirement Analysis Document

BOOK STORE MANAGEMENT SYSTEM

1. Objective & Scope

This document specifies the software requirements for the full-stack Book Store Application. The system features a modern decoupled architecture using an Express/Node.js backend tier and a reactive client built with Vite, React, and Tailwind CSS. The app features state-driven catalog tracking, order processing pipelines, and administrative analytics dashboards.

2. Functional Requirements (FR)

2.1 User Authentication Module (/api/auth)

- **Firebase Integration:** Secure client-side registration and login handled seamlessly via Firebase SDK wrappers.
- **Session Security:** Route protection and authentication state binding across navigation actions using persistent user contexts.

2.2 Catalog & Discovery Module (/api/books)

- **Book Carousel Display:** Dynamic asset presentation utilizing `Swiper.js` layout grids to slide through trending genres.
- **Inventory Search:** Content filtering and index retrieval matching specific metadata models fields.

2.3 Checkout & Order Module (/api/orders)

- **Persistent State Cart:** Localized cart state processing powered by Redux Toolkit slice actions to handle additions, updates, and subtractions.
- **Transactional Delivery:** Formalized checkout operations requiring customer shipping address profiles and logging pending records.

2.4 Admin Analytics Dashboard (/api/admin)

- **KPI Visualization:** Real-time reporting showing sales trends and total order volumes mapped out visually through responsive `Chart.js` graphics.

3. Non-Functional Requirements (NFR)

- **Cross-Origin Resource Sharing (CORS):** Strict domain whitelisting configured to securely bridge credentials between Vercel deployment environments and backend servers.
- **UI Responsiveness:** Smooth screen adaptations across standard layouts using fluid utility configurations from Tailwind CSS.